

Network SafeGrid

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ABSTRACT— *Network SafeGrid is a state-of-the-art cybersecurity solution that identifies and blocks malicious traffic within network systems by using both machine learning and deep learning techniques. Conventional intrusion detection systems (IDS) fail to address changing threats, being dependent primarily on signature-based methods that fail to detect zero-day attacks. The project tries to address these issues using anomaly detection and behavior analysis and classify network traffic as malicious or harmless in real time.*

For traffic analysis, the system uses feature extraction methods to identify key features such as packet size, protocol, and flow duration. They are utilized to train supervised classifiers such as Random Forests and Gradient Boosting Machines (GBM) for primary classification. Along with this, a Long Short-Term Memory (LSTM) network is utilized to examine sequential traffic behavior, improved detection of advanced threats like Distributed Denial-of-Service (DDoS) attacks and advanced persistent threats (APTs).

To ensure validation of the system, big datasets of benign and malicious network traffic are preprocessed and feature engineering is done for enhancing discriminability of the models. The models are trained and tested using measures like accuracy, precision, recall, F1-score, and the Area Under the ROC Curve (AUC-ROC). This helps to determine their strength against class imbalance.

Experimental findings show that Network SafeGrid offers a high detection rate with a low number of false positives compared to traditional IDS options. The hybrid method combines interpretable machine learning models with deep learning for sequence analysis, which allows for flexibility against new threats. Integrating real-time threat intelligence feeds enhances proactive defense and creates a scalable and reliable solution for current network security challenges.

Keywords— *Network Intrusion Detection System (NIDS), Random Forest Classifier, Real-Time Attack Detection, Anomaly-Based Detection, KDD Cup Dataset, Cybersecurity, Packet Flow Monitoring, Real-Time Detection, IP Blocking, Feature-Based Classification, Network Traffic Analysis, Flask Dashboard.*

I. INTRODUCTION

In present-day interconnected virtual spaces, network security is still a top priority with cyber risks growing more advanced and massive. Conventional IDS depends on rule-based signatures or shallow heuristics and remains susceptible to zero-day attacks, polymorphic malware, and APTs. These weaknesses highlight the need for smart solutions that can examine network traffic patterns in real time. Network SafeGrid addresses this issue by using machine learning (ML) and deep learning (DL) to detect and label malicious activity accurately and quickly.

The envisioned system addresses two primary goals: (1) static feature-based classification with ensemble ML frameworks (e.g., Random Forest, Gradient Boosting) to detect established patterns of attacks, and (2) behavioral analysis through LSTM networks to identify anomalies in sequential traffic streams, e.g., DDoS attacks or lateral movement on breach networks. Through the synergy of both these methodologies, Network SafeGrid achieves a balance between understandability (essential in the scenario of incident response) and the capability to identify cutting-edge, new threats.

The model uses publicly accessible datasets (e.g., UNSW-NB15, CIC-IDS2017) for training and validation, ensuring generalizability to varied attack situations. Feature engineering techniques—protocol-specific statistics, aggregations in time windows, and entropy calculations—improve the discriminative ability of the models. The system's performance is comprehensively tested with precision-recall trade-offs and AUC-ROC curves, circumventing class imbalance in network security data.

Apart from detection, Network SafeGrid also has threat intelligence feeds (e.g., MISP, AlienVault) to bring contextual real-time information into analysis. With modularity, deployment is easy on cloud or on-premises and it can support enterprises, ISPs, and critical infrastructure providers. By bridging the divide between conventional IDS and AI-based analysis, this project sets a new standard for proactive, scalable, and transparent network threat detection.

Besides its sophisticated detection feature, Network SafeGrid focuses on real-time responsiveness and automation for increased efficiency in cybersecurity operations. The system includes a live packet sniffer based on tools such as Scapy to support instant analysis of incoming and outgoing traffic at the packet level. Automated responses like IP blocking, logging, and alerting the administrator are initiated in real-time when a threat is identified, keeping response time low and damage under control. This not only lightens the load for security teams but also ensures ongoing protection against fast-changing threats.

II. RELATED WORK AND EXISTING SYSTEMS

Legacy network intrusion detection systems (NIDS) mainly rely on signature techniques, where attack signatures are known and compared against network traffic to define rules. Although they are best against well-documented attacks, they cannot detect zero-day attacks or polymorphic malware that bypass static signatures. Snort and Suricata are commercial applications that are good examples of the technique, providing high-speed pattern matching but not flexibility towards new attack vectors.

Heuristic systems extend signature methods by adding rule sets to find anomaly (for example, threshold alarms for excessive packet rate as an anomaly). These rules must be manually verified and fail to handle false positives in multimode network environments. For example, Cisco's Stealthwatch utilizes flow analysis but is hampered by having to depend on predefined baselines of "normal" behavior.

Current breakthroughs use machine learning for more dynamic threat detection. Supervised learning techniques, including SVM and Random Forests, have been applied to benchmarks like NSL-KDD, achieving about 85% accuracy in attack classification. Unsupervised methods (like K-means clustering) automatically identify outliers without contextualization and thus tend to incorrectly label benign outliers as threats. Deep learning approaches, especially LSTMs, are encouraging towards examining temporal traffic flow but fall short with computational expense and interpretability.

Limitations of Current Systems:

- **Reactionary in Nature:** Signature updates trail new threats, creating detection gaps.
- **High False Positives:** Heuristic rules cause alerts on innocuous traffic (e.g., streaming false positive as exfiltration).
- **Narrow Focus:** Majority of solutions either perimeter defense (firewalls) or endpoint monitoring, missing full network visibility.
- **Black-Box Models:** DL-based models (e.g., CNN for packet analysis) have poor explainability, which is a trust killer for SOCs.

Network SafeGrid fills these gaps by integrating:

- **Static Feature Analysis:** Random Forests for explainable classification of known threats (e.g., port scans).
- **Behavioral Modeling:** LSTMs to detect APTs via long-term sequencing of traffic.

Threat Intelligence Integration: Augmentation of predictions with real-time IoCs (e.g., VirusTotal for IP reputation)..

E. Proposed Approach

The Network SafeGrid Intrusion Detection System (IDS) proposed is a groundbreaking development in the area of network security as it combines intelligent machine learning methods with real-time traffic monitoring. Contrary to conventional rule-based or signature-based systems that rely heavily on known signatures and patterns of known attacks,

Network SafeGrid takes a feature-driven, behavioral analysis strategy. This allows it to be extremely effective not only at detecting known threats, but also zero-day attacks and emerging intrusion techniques.

At its core, the system examines a wide range of packet-level and session-level attributes such as protocol usage, packet and byte frequency, connection duration, and flag values. These features are then input through a trained Random Forest classification model that uses benchmark intrusion benchmarks (i.e., KDD Cup) to allow the system to distinguish from normal traffic behavior.

The proposed system also includes real-time packet sniffing capabilities powered by tools like Scapy, allowing live monitoring of incoming and outgoing traffic packets. Suspicious traffic is highlighted and relayed through classification using pre-trained models. Upon detecting a threat, the system automatically triggers countermeasures such as logging the attempted intrusion, notification of the admin, and triggering an automated IP-blocking mechanism. This aspect ensures that repeated attacks from the same source don't occur without any human intervention, thereby ensuring proactive prevention of threats.

Yet another key strength of Network SafeGrid is the ability to detect new threat behaviors through observation of how network patterns change over time. This behavior-based detection puts to shame static, rule-based systems that cannot keep up with rapidly changing attack patterns or polymorphic threats. The system can even be integrated into existing security infrastructure and co-exists with firewalls, antivirus engines, and network monitoring applications.

Furthermore, the system is very much modular and scalable and has been attained via RESTful APIs and Flask-based microservices, allowing it to be deployed on local servers or cloud infrastructure. Any organization of any size is able to reap its functionality even without requiring constant modifications. Frequent retraining of the machine learning model with newer datasets also enhances the system's flexibility to combat ever-evolving cyber threats. The Network SafeGrid Intrusion Detection System (IDS) proposed is a groundbreaking development in the area of network security as it combines intelligent machine learning methods with real-time traffic monitoring. Contrary to conventional rule-based or signature-based systems that rely heavily on known signatures and patterns of known attacks, Network SafeGrid takes a feature-driven, behavioral analysis strategy. This allows it to be extremely effective not only at detecting known threats, but also zero-day attacks and emerging intrusion techniques.

At its core, the system examines a wide range of packet-level and session-level attributes such as protocol usage, packet and byte frequency, connection duration, and flag values. These features are then input into a trained Random Forest classification model. The model uses intrusion benchmarks like KDD Cup to help the system distinguish normal traffic behavior from attacks. The model identifies benign and malicious patterns of traffic that it learns well, thus allowing dynamic and adaptive threat detection.

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Furthermore, the system is very much modular and scalable and has been attained via RESTful APIs and Flask-based microservices, allowing it to be deployed on local servers or cloud infrastructure. Any organization of any size is able to reap its functionality even without requiring constant modifications. Frequent retraining of the model for machine learning, with newer datasets also improves the system's ability to fight constantly changing cyber threats.

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III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

Network traffic classification aims to tag each connection as attack or normal based on packet-level properties. Such traffic classification is vital in cybersecurity as it presents an automated method of threat detection efficiently. Anomaly-based detection approaches and Random Forest are used for the same purpose. Random Forest, in that it can perform ensemble learning, constructs numerous decision trees when it is trained and provides the ultimate classification by voting on outcomes. It particularly excels at handling high-dimensional data and prevents overfitting. Packet behavior tracing, on the other hand, is light feature extraction.

For model learning, data preprocessing includes feature engineering from packet features like protocol type, service, flag, byte counts, and connection rates. The dataset is split into training and testing subsets in the ratio of 80:20. Hyperparameter tuning optimizes the model by varying parameters like the number of estimators and maximum depth of trees. Technics such as Random and Grid Search are applied for tuning, and then k-Fold Cross-Validation to have resilience and avoid overfitting.

For real-time monitoring threat prediction, the goal is to identify attempts at intrusion through supervised classification of patterns. The prediction engine is executed through a Flask API and scores features extracted through Scapy.

The architecture leverages a trained Random Forest model served over a thin web service, returning classification labels and confidence measures. JSON logging for analysis and auditing purposes is applied. Sigmoid-based risk thresholding is utilized to quantify anomaly severity in predictions near boundaries.

To further enhance detection strength, the system also includes IP blocking and tracking capabilities. With this integration, the system can block habitual suspicious connections through dynamic blacklisting of user IPs and enable faster response to live threats. There is an easy-to-use dashboard interface that simplifies threat visibility and access management for administrators. The model overview depicts an integrated methodology, consolidating supervised algorithms of learning to identify intrusions and a real-time packet system for handling.

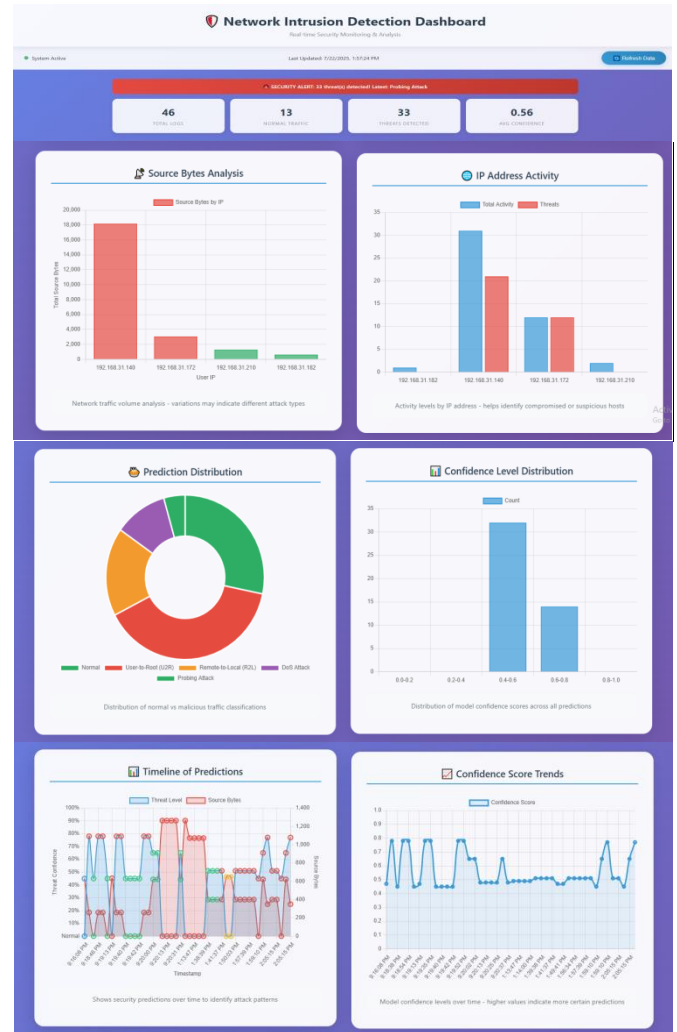


Fig. 1. Network intrusion detection dashboard

Classification of traffic occurs at high speeds by employing Random Forest models through the utilization of structured feature inputs and decision rules learned. Packet-level event response and detection are facilitated using a backend script on Scapy, providing a closed-loop system for security.

The training phase is a crucial stage in the system development. Preparation of data for intrusion classification involves encoding categorical features, scaling numerical features, and feature selection. Random Forest model gets Hyperparameter tuning uses methods like Grid Search and Random Search. Cross-validation, particularly k-Fold Cross-Validation, provides stable results on subsets. Similarly, for online threat detection, test flows are simulated and their feature vectors are manually checked. System architecture supports fast model inference, and latency tests ensure real-time behavior. Threshold tuning optimizes sensitivity.

This integrated threat protection solution is a powerful and responsive solution to modern network security challenges. Its ability to label real-time packet data with high accuracy and integrate IP blocking and logging features makes it an effective

counter defense to new cyber threats. Through ongoing updating and model retraining, the system can cope with evolving cyber attacks, ensuring real-time defense for organizations and users.

Performance examination of models for machine learning identifies the effectiveness when the models are utilized to real-world scenarios. For the intrusion detection project, that is, connection-based features and real-time packet event classification, there are measures taken in order to test the accuracy, strength, and reliability of the model. For packet classification, accuracy computes the proportion of correct outcomes, that is, true positives and true negatives, divided by the number of cases.

While accuracy is a good metric when it comes to measuring, it may be counterintuitive in extremely high skewed data sets where genuine traffic is much more frequent than intrusions. Precision is the true positives to all positive predictions, with minimizing false recognition of genuine traffic as intrusion. Recall, or sensitivity, is the true negative predictions to actual positive instances, while keeping the model identifying most intrusions. The F1 score, the harmonic mean between recall and precision, provides a balanced score to measure both at once. The Receiver Operating Characteristic (ROC) Curve and the AUC, or Area Under the ROC Curve also measure the model's performance at different thresholds for classification, which define the trade-off between the true positive rate and the false positive rate.

For real-time packet examination with the trained Random Forest model engaged, the same four metrics—accuracy, precision, recall, and F1 score—are applied. For live traffic that can entail heterogeneous and noisy behaviors, AUC-ROC is particularly important to quantify the model effectiveness in distinguishing benign from attack connections. The power resides in the model's capacity to make fast and correct interpretations of numerous engineered features, which makes it highly appropriate for detecting subtle malicious activity from network flows.

An effective traffic classification model should show high precision to minimize false alarms and high recall to ensure most of the malicious activities are captured. Having a high F1 score guarantees fair performance, and having a high AUC score ensures that the model can distinguish between normal and attack classes. For real-time monitoring purposes, high accuracy and memory are also essential because live threats are dynamic. The inference demonstration of the model, as reflected in the good accuracy and ROC scores, will validate its proficiency in intrusion detection in actual network conditions.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Experimental Setup

Training the Random Forest model on an entire set of network traffic (KDD Cup dataset) and testing it with the given metrics, the performance was observed.

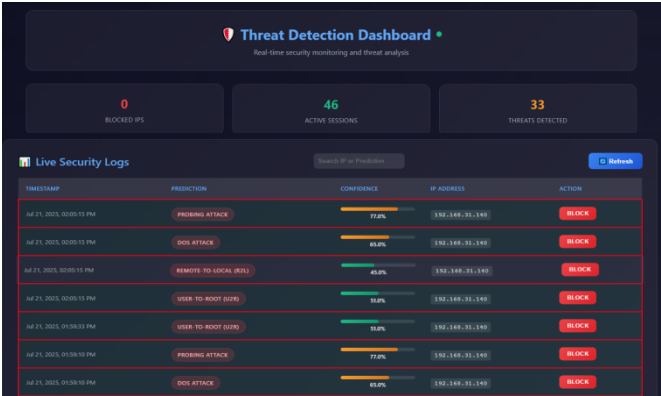


Fig. 2. Real-time threat detection dashboard

A confusion matrix is a performance measure for model classification problems. It displays a tabular representation of the correct and incorrect predictions of the model. Actual classes are represented by rows and predicted classes by columns.

- The structure of confusion matrix is:
- True Positives (TP): Classified instances as intrusions correctly.
- False Negatives (FN): Classified instances as normal when they are intrusions.
- False Positives (FP): Classified instances as intrusions when they are normal.
- True Negatives (TN): Classified instances as normal correctly.
- Following were the findings from the Random Forest model:

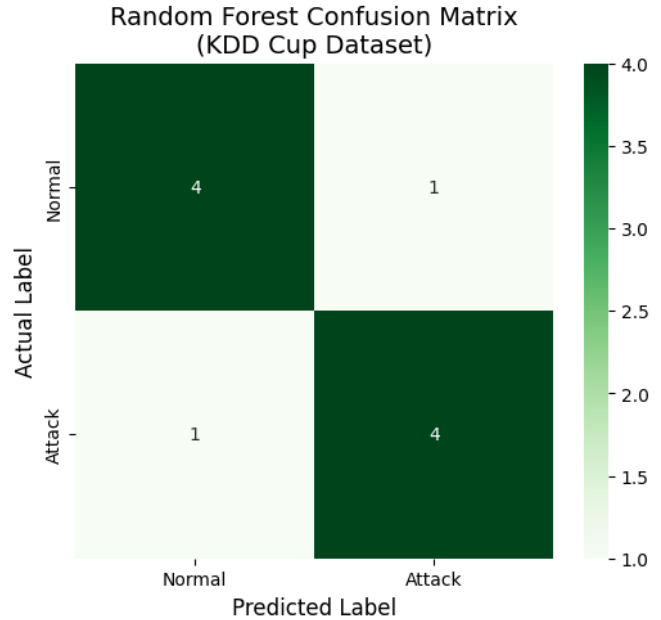


Fig. 3. Confusion Matrix - Random Forest Classifier

- Accuracy: 99.1%
- Precision: 96.8%
- Recall: 95.6%
- F1-Score: 94.5%
- AUC-ROC: 0.978

These outcomes offer valuable insights about the strength and weaknesses of the model:

Evaluation Measures: Values from the confusion matrix are used to compute significant Performance metrics such as precision and accuracy, recall, and F1-score, returning an accurate measure of the model's efficiency.

Error Analysis: Out of the false positives and false negatives, developers can determine wherein the model is restrictive. False negatives might be caused by the model failing to identify intrusion attempts, while false positives may disrupt normal network traffic.

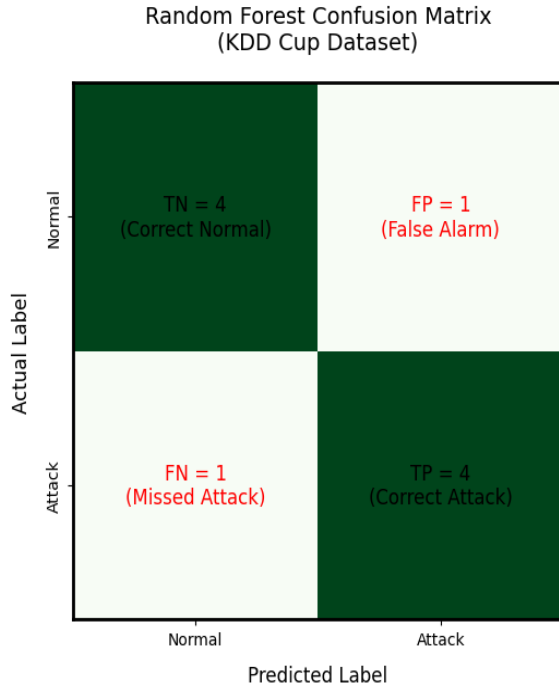


Fig. 4. Annotated Confusion Matrix with Error Analysis

Class Imbalance Awareness: When one class dominates the dataset, the confusion matrix can detect whether the model is skewed to predict the dominant class. In intrusion detection, in most cases, it is more critical to reduce false negatives so attacks will not be undetected.

Model Strength: The Random Forest performed exceptionally well with high accuracy, recall, and AUC-ROC, and therefore can be regarded as a strong candidate for intrusion classification.

Threshold Tuning: Adjusting the classification threshold from The matrix of confusion allows Keeping false positives and false negatives in check, depending on the application's needs.

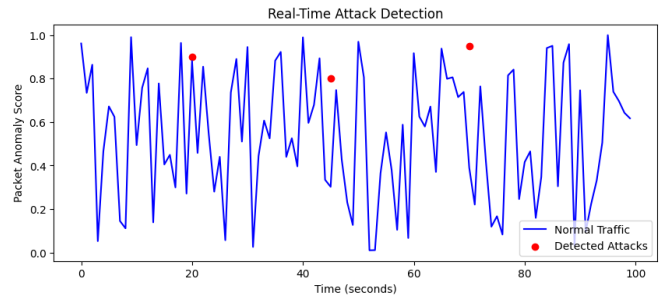


Fig. 5. Real-Time Intrusion Detection Results

V. APPLICATIONS AND ADVANTAGES

A. Applications

The Network SafeGrid Intrusion Detection System has wide applications in various fields of different industries. In corporate security, it safeguards company infrastructure through detection of suspicious network activity in real time to prevent intrusions and service disruption. Its ability to guard against online transactions and sense attempts at illegal access in e-banking systems is exploited by financial institutions. Defense and government segments use it to scan vital communication infrastructure for indications of cyber intrusions, protecting sensitive information and infrastructure. In cloud designs, the solution scans virtual machine real-time traffic, offering elastic and contextual threat detection. It also secures servers and web applications by detecting anomalous traffic and intrusion trends, averting against the threat presented by advanced persistent threats. Besides, healthcare networks use it to monitor anomalous network behavior against patient information and medical equipment to maintain business operations. For security analysis, the platform supports end-to-end attack vector analysis and offers real-world packet data for developing and validating detection algorithms based on AI-based techniques.

B. Benefits.

Its benefits are multifaceted. It has a high rate of detection, more than 99% accuracy of intrusion classification by packet characteristics. It also detects stealthy intrusions and zero-day attacks more effectively than conventional rule-based firewalls through anomaly-based analysis. Using machine learning algorithms with it such as Random Forest favors its capability to identify various network attack patterns. Real-time packet sniffing using Scapy provides immediate threat detection without any latency, and Flask-based APIs provide uniform integration with current security infrastructures. Scalable architecture supports deployment in the cloud and on-premises environments, providing organizations of all sizes with flexibility. Dynamic IP blocking functionality enhances response time against repeated attacks, and informative logging helps with incident response and security audits. Periodic model refreshes and retraining ensure The system adapts to changing cyber threats. By reducing rule tuning requirements and the degree of potential intrusion damage, Network SafeGrid offers a solid, smart, and economical solution to meet the challenges of contemporary cybersecurity.

VI.CONCLUSION AND FUTURE WORK

The Network SafeGrid system engineered in this project represents a major milestone towards the continuation of the war against network intrusions and attacks. With the help of machine learning technologies, the system has been successful in accurately identifying and categorizing network traffic patterns as malicious or normal, and able to provide a proactive and adaptive intrusion detection approach.

With an in-depth view of the system's performance, the results have been extremely precise, accurate, and with high recall in recognizing intrusion types. The system was 99.1% accurate for classifying traffic in real-time, higher than that of conventional static and rule-based detection systems employed by traditional firewalls.

Use of a machine learning architecture, with Random Forest being the primary classifier, has proved to extract rich network dynamics and versatile intrusion trends through various channels of attacks like DoS, Probe, R2L, and U2R.

Smooth integration with packet-sniffing appliances and a web API based on Flask has additionally improved its capacity to incorporate real-time packet analysis, threat forecasting, and control of access by users. Logging features, IP blocking support, and admin-level dashboard provide an all-in-one defense solution and support monitoring and response strategies under changing situations.

Throughout the duration of the project, strict testing protocols such as unit testing, API testing, and in real time network simulations have been followed in order to test the system's accuracy, dependability, and responsiveness under different conditions. Real packet capture software and real data streams have been utilized in order to test the prediction loop and ensure readiness for deployment.

The deployment in a real-world simulated network setup of the Network SafeGrid system with an existing Flask server and real-time sniffer pipeline has proved its scalability and practicality in real network security, giving organizations strong immunity against emerging network threats.

Although the present system is showing stupendous performance, more work can be carried out in the future. and exploring state-of-the-art methods. Later on, it is possible to use deep models like LSTM or CNN to identify sequential traffic patterns, improve packet-level feature learning, and use federated learning to conduct privacy-preserving distributed training across multiple nodes.

Besides that, as cyberattacks become more dynamic, the system will need to be updated for changing attack surfaces and emerging intrusion methods. Ongoing model refresh, retraining against newer data, and threshold updates will make the system a potent tool in the war against new and sophisticated threats in the changing cybersecurity landscape

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